



The University of Lahore

Department of CS & IT

Course: Advanced Software Engineering (Adv SE)

Assignment # 1 10marks Due Date: 13-03-2026

Assignment: Real-World Problem Discovery and Product Inception

Objective:

You must look at your daily life, university campus, local community, or a specific industry you are familiar with, and identify a genuine, frustrating inefficiency or pain point. Your task is to validate that this problem exists, define a Minimum Viable Product (MVP) to solve it, and translate that solution into an actionable Software Engineering plan.

Part 1: Problem Discovery & Validation (3marks)

Engineers often invent solutions for problems nobody actually has. Before planning the software, you must prove the problem is real.

The Task: Identify a broken process, an underserved need, or a manual task that is ripe for automation. Interview at least three people affected by this problem (stakeholders).

Deliverables:

1. **The Problem Statement:** A concise, 2-3 sentence description of the current state, the pain it causes, and the impact of that pain.

(Example: "Currently, university clubs manage their budgets across disjointed WhatsApp groups and paper receipts, resulting in lost funds and a 3-week delay in reimbursement approvals.")

2. **Stakeholder Personas:** Create 2 distinct user personas based on your interviews. Include their goals, technical literacy, and primary frustrations.
3. **Validation Evidence:** A brief summary (1 paragraph) of what you learned from your stakeholder interviews. Did their actual problems match your initial assumptions?

Part 2: The Lean Solution & Scope Definition(3marks)

Now that the problem is validated, propose a software solution. The focus here is on constraint: what is the absolute minimum you can build to solve the core problem?

- **The Task:** Define the boundaries of your Minimum Viable Product (MVP).
- **Deliverables:**
 - **Lean Canvas:** Fill out a 1-page Lean Canvas focusing on the Problem, Solution, Unique Value Proposition, and Customer Segments.
 - **In-Scope vs. Out-of-Scope Table:** Clearly list 3-5 core features that *must* be in the MVP, and 3-5 "nice-to-have" features that are strictly deferred to future versions.

Part 3: Agile Requirements Translation(3marks)

Translate your proposed solution into artifacts that an engineering team can actually use to start working.

- **The Task:** Break down the MVP into manageable Agile components.
- **Deliverables:**
 - **System Epics:** Define 3 high-level Epics
(e.g., "User Authentication & Onboarding," "Receipt Upload & OCR Processing").
 - **User Stories:** Write 5 critical User Stories using the standard format: "As a [persona], I want to [action] so that [value/reason]."
 - **Acceptance Criteria:** For two of your User Stories, provide detailed acceptance criteria defining exactly what conditions must be met for the story to be considered "done."

Part 4: Feasibility and Risk Assessment(1mark)

Real-world planning requires anticipating constraints.

- **The Task:** Analyze the technical and non-technical risks associated with your proposed system.
- **Deliverables:**
 - **Risk Matrix:** Identify 3 major risks (Consider categories like Data Privacy/Security, Third-Party API reliance, User Adoption, or Technical Complexity).
 - **Mitigation Strategy:** Write a short paragraph for your highest-impact risk explaining how your architecture or project plan will mitigate it.